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# Exploring Vision Transformers for Weakly Supervised Semantic Segmentation for Lung Adenocarcinoma Histopathology Images

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## Abstract

It is a challenge to perform weakly supervised semantic segmentation (WSSS), especially when pixel-level prediction is being supervised by image-level labels. A Class Activation Map (CAM) is typically created to offer pixel-level pseudo labels to fill the space between them. Convolutional neural networks' CAMs experience partial activation or activation of only the most discriminative regions. On the other hand, transformer-based approaches are very good at investigating global context with long-range dependency modelling, perhaps resolving the "partial activation" problem. In this study, we evaluate the efficacy of several transformer networks for the task and present the first transformer-based WSSS technique for histopathology pictures. Unlike other CNN-based techniques, we train a segmentation model for semantic segmentation using high-quality pseudo segmentation masks created by Vision Transformer (ViT), a classifier. Additionally, we adapt the Pixel-Adaptive Refinement module, which incorporates low-level picture appearance information to refine the pseudo labels, to assure accuracy and local consistency. Extensive testing on the WSSS4LUAD dataset showed that this method can successfully segment histopathological images using only image-level labels. The code is available on github

## 1 Introduction

Lung cancer is the leading cause of cancer death worldwide, with the most common sub-type being lung adenocarcinoma (LUAD) Sung et al. [2021] UK [2022] Thai et al. [2021]. Histopathology examination of a tissue section stained with hematoxylin and eosin (H&E) remains the gold standard in the diagnosis of LUAD. It provides extensive information on the tumour micro-environment from various perspectives, such as tissue morphology, spatial arrangement at the tissue or cell level, nuclear atypia, and so on Hanahan [2022]. Experienced pathologists can assess tumour progression by reading the patterns of interaction between the tumour and its micro-environment Bremnes et al. [2011]. However, it is dependent on the experience of the pathologists and has limited inter-observer agreement Yu et al. [2018]. It is strongly advised to translate the visual language to the computer language for an objective, quantitative, and precise analysis of histopathology sections Han et al. [2022b].

One of the most crucial upstream tasks in digital pathology is semantic segmentation, which specifies the tissue type for every single pixel and enables quantitative analysis for the prediction of cancer results Srinidhi et al. [2021]. The performance of semantic segmentation Cheng et al. [2020] Li et al.

[2020] Chen et al. [2018] Zoph et al. [2020] algorithms has improved rapidly in recent years as a result of the advancement of deep learning. However, to achieve superior performance, most recent deep-learning-based algorithms have become reliant on large labelled datasets. OCNet Yuan et al. [2020], DANet Fu et al. [2019], HRNet Wang et al. [2020b], Deeplab3+ Chen et al. [2018] are all fully-supervised semantic segmentation (FSS) methods. Although fully-supervised methods achieve state-of-the-art performance, obtaining dense pixel-level labelled datasets to train deep learning networks is one of the major challenges for semantic segmentation research. This limitation in labelled datasets is much more apparent in semantic segmentation than in other vision tasks such as image classification or object detection.

To address this issue, studies on weakly supervised semantic segmentation (WSSS) have used labels with limited information, such as image-level labels Ahn et al. [2019] Chang et al. [2020] Han et al. [2022b], scribbles Vernaza and Chandraker [2017], and bounding boxes Papandreou et al. [2015], to generate dense pixel-level labels. Using only image-level classification labels, in particular, is the most actively researched approach in WSSS because image-level labels are much more accessible than other labels. The average time for MS COCO annotation, which is 150 times faster than image-level annotations, was 4.1 seconds for categorising each image and 10.1 minutes for labelling each image by individual pixels Lin et al. [2014]. As they are trained with the most informative supervisory data feasible, fully-supervised algorithms are regarded by WSSS researchers as having the "upper bound" in performance (assuming the annotations are reasonably numerous and accurate) Kwak et al. [2017] Kervadec et al. [2019]. At the time of writing, the best fully-supervised method, EfficientNet-L2+NAS-FPN Zoph et al. [2020], achieved a 90.5 per cent mIoU on the PASCAL VOC2012 test set Everingham et al. [2012], which is significantly higher than the current best weakly-supervised method, RSCM Jo et al. [2022], which achieved a 73.2 per cent mIoU.

WSSS methods are typically developed for natural scene image benchmark datasets like PASCAL VOC2012, with growing interest in research on applying them to other image domains like histopathology images Han et al. [2022b] Hosseini et al. [2019]. While we might expect re-trained WSSS methods to perform similarly for different image domains, these images differ significantly from natural scene images. Natural scene images have finer-grained visual information (low intra-class variation and high inter-class variation), whereas satellite and histopathology images have coarser-grained objects (high intra-class variation and high inter-class variation) Xie et al. [2019]. Furthermore, unlike with natural scene images, the boundaries between objects are frequently imprecise, and even professionals disagree when labelling histology Xu et al. [2017] Han et al. [2022b] and satellite images Mnih and Hinton [2010]. Chan et al. [2021] implemented the state-of-the-art (SOTA) methods at the time in various domains (natural scene, satellite images and histopathology) and observed that the methods developed for the natural scene (i.e. SEC Kolesnikov and Lampert [2016] and DSRG Huang et al. [2018]) and histopathology images (i.e. HistoSegNet Chan et al. [2019]) indeed performed best in their intended domains. Furthermore, they showed that most methods perform moderately well for satellite images while many methods performed poorly on datasets they were not designed to solve (such as HistoSegNet in VOC2012, SEC and DSRG in ADP Hosseini et al. [2019]), although IRNet seemed to be most robust overall to dataset choice.

Prevailing WSSS with image-level labels are mostly based on convolutional neural networks, which have a limited receptive field and fail to explore global-level relations. Recent research on Transformers has shown significant breakthroughs in computer vision tasks (Vision Transformers - ViT). The application in WSSS has only just started gaining '*attention*' as they have been proven to extract global context information which is vital for semantic segmentation Liu et al. [2022] Sun et al. [2021b] Ru et al. [2022]. Although vision transformers have been explored for classification and WSSS tasks, their application on WSSS for histopathological images is yet to be explored, where the retention of global context is potentially much more important, as, unlike natural scene images, object boundaries in histopathological images are rather ambiguous. Our contribution can be summarised as follows:

- With the ability to retain global context, we explored the use of various vision transformer networks and attention map generation in ViTs from recent works in natural scene images. We explored the effect of various augmentations on the classification accuracy from the models as well as on the quality of activation maps generated.
- We adapted PAMR from Ru et al. [2022] to refine the attention maps which were then further constrained by thresholding to retain the most important sections of the activation map. These refined activation maps were then used as pseudo-labels for DeeplabV3+Chen et al. [2018] for training and the segmentation model performance was evaluated against the validation and test dataset.
- We explored the application of recent end-to-end WSSS with transformer developed for PASCAL VOC and COCO datasets where the models have produced state-of-the-art results Sun et al. [2021b] Ru et al. [2022].

## 2 Related Work

### 2.1 Histopathological Image Segmentation:

With the advancements in deep learning techniques, computational pathology has seen significant breakthroughs Srinidhi et al. [2021] Deng et al. [2020]. Histopathology image segmentation is one of the key processes in biomedical imaging and analysis, with state-of-the-art performance in the fields of localization and/or segmentation of tissues Van Rijthoven et al. [2021], glands, nuclei Zhao et al. [2020], tumors Han et al. [2022a], etc. With the data-driven nature of these methodologies, sufficient labelled data is required which is traditionally prepared manually and as domain knowledge is vital in the identification of objects in biomedical imaging, it is an expensive and time-consuming task.

Four deep learning techniques are utilised in Swiderska-Chadaj et al. [2019] to automatically identify lymphocytes in images of histopathology. The four deep learning techniques are (1) FCN-based patch classification; (2) U-net for cell segmentation; (3) YOLO-based detection; and (4) Locality Sensitive Method for prediction. The dataset comprises 37 training photos, 6 validation images, and 40 testing images. The images depict breast cancer, colon cancer, and prostate cancer. Last but not least, U-net turns in the best result with an F1 score of 0.78. Additionally, compared to pathologists ( $k = 0.64$ ), U-net and manual evaluation had a better level of agreement ( $k = 0.72$ ). A fully convolutional neural network (FCNN) is employed in Bándi et al. [2019] to distinguish between tissue and background regions to prevent inefficient scanning in void background areas. There are seven convolutional layers; the ReLU activation function is used in the first six layers, and the softmax activation function is used in the final layer. The segmentation performance of FCNN and U-net is comparable. This collection contains 100 WSIs for organs divided into 10 categories, each with 10 WSIs. There are 10 lymph node WSIs stained with CK8-18 and 10 with H&E. In a network that does not care about resolution, FCNN produces Dice scores ranging from 0.97 to 0.98, and on eight test images, it earns a Dice score of 0.97 and sensitivity of 0.98.

### 2.2 Weakly Supervised Semantic Segmentation (WSSS):

On account of the niche domain knowledge required for the correct identification of elements in biomedical imaging, significant efforts have been put into weakly supervised semantic segmentation for obtaining pixel-level labels from image-level labels Han et al. [2022a]. Extensive research has been performed in WSSS to obtain pixel-level labels from image-level labels and is traditionally implemented for natural scene image datasets such as PASCAL VOC and COCO. WSSS has mostly been performed in multiple stages, where a classification network is trained from which initial pixel level pseudo-labels are generated using class activation maps (CAMs) Selvaraju et al. [2017] Chattopadhyay et al. [2018] Wang et al. [2020a] for training semantic segmentation models. To address the drawback of the model focusing solely on the most discriminative features, Sun et al. [2021a] Zhang et al. [2021a] Han et al. [2022a] erase the most discriminative regions from the CAMs, forcing the network to focus more on the complete object regions. Consecutively, Huang et al. [2018] apply a region growing algorithm following a classification network for refining class activation maps for pseudo-labels generation for semantic segmentation.

Research in the development of end-to-end model training for WSSS is gaining more traction, however, due to the extremely limited supervision training of such models has proven to be quite challenging. Papandreou et al. [2015] have adopted an expectation maximization framework using intermediate predictions as segmentation labels. Yun et al. [2019] have presented an end-to-end framework to train classification and segmentation models simultaneously with integration to obtain a reliable pseudo-segmentation mask. RRM employs CRF Krähenbühl and Koltun [2011] over CAMs to produce refined labels for segmentation. They also introduce an auxiliary regularization loss Tang et al. [2018] to ensure consistency between segmentation mask and low-level image features. With all the aforementioned research, the end-to-end WSSS is still a developing field and has clear performance gaps from multi-step methods.

### 2.3 Vision Transformers in WSSS:

Dosovitskiy et al. [2020] proposed vision transformers (ViTs) for the first time to apply pure transformer architecture developed for natural language processing to computer vision tasks achieving breakthrough performance in the computer vision classification benchmark. Later variants of ViT have also shown significant improvements in image classification tasks Touvron et al. [2021] Liu et al. [2021] and other computer vision tasks such as image segmentation Zheng et al. [2021], etc. With the spectacular performance of vision transformers in image classification and other computer vision tasks, researchers have taken note of the capability of vision transformers to retain global information and have shown breakthrough performance for natural scene datasets such as PASCAL VOC and COCO. Liu et al. [2022] proposed a transformer-based WSSS framework and introduced Deep Taylor Decomposition for CAM generation, followed by soft erasure and efficient potential object mining (EPOM) to refine CAMs and generate high-quality pseudo-labels. Without any ‘bells and whistles,’

the authors have observed state-of-the-art 70.5% mIoU on the PASCAL VOC dataset Liu et al. [2022]. Sun et al. [2021b] proposed a transformer based WSSS and introduced Gradient-weighted Element-wise Transformer attention Map (GETAM) Sun et al. [2021b]. They further proposed an activation-aware label completion module for generating high-quality refined pseudo-labels and achieved state-of-the-art segmentation accuracy not only in end-to-end methods but also comparable performance to most multi-stage methods.

## 2.4 Semantic Segmentation:

Following the work of Long et al. [2015], fully convolutional networks (FCN) have garnered the most attention in recent years for semantic segmentation. FCNs only use convolutional layers that are passed to a classification layer; they do not use any fully-connected layers. Dramatically lowering the number of parameters in the network and preserving geographical information from the input, improves computational efficiency. With U-Net Ronneberger et al. [2015] and SegNet Badrinarayanan et al. [2017] all adopting the same concept, whereby the first half of the network is a mirror image of the second half, albeit, with distinct systematic details, encoder-decoder type networks achieve excellent results. By preserving the indices of the maximum values on the max-pooling operation, SegNet Badrinarayanan et al. [2017] presents a computationally quick and memory-efficient network. This is subsequently used during up-sampling in the decoder component of the network. By using skip connections, U-Net makes it possible to use data from the network’s encoder portion throughout the decoding process.

Subsequently, the DeepLab family of architectures Chen et al. [2018] Chen et al. [2014] Chen et al. [2017] pushed the limits of semantic segmentation using atrous convolutions with Chen et al. [2017] successfully creating the Atrous Spatial Pooling Pyramid (ASPP) module to handle objects of various scales in the same image. The foundation of ASPP is the simultaneous application of dilated convolutions with various atrous rates and the concatenation of the resulting feature maps. In essence, this mixes many frames of vision to address the problem of objects of various scales. With each network iteration, the DeepLab family has produced cutting-edge results on benchmark datasets.

## 3 Methodology

Manually annotating histopathological images at the pixel level takes a lot of time and requires subject-matter knowledge or skilled pathologists. The objective of this study is to develop a reliable weakly supervised semantic segmentation technique for lung adenocarcinoma histopathology pictures. We suggest using CAMs from vision transformers to generate initial pseudo-labels. Next, we use pixel adaptive refinement (PAR) Ru et al. [2022] to make sure that the pseudo labels are consistent locally. A semantic segmentation model was then trained using these labels as the starting point. This is the first use of transformers for WSSS in histopathology images as of the time of writing. figure 1 demonstrates the systematic design of our proposed approach.

### 3.1 Vision Transformers

The vision transformer served as the classification network in our method, which was trained on the LUAD dataset, as depicted in figure 1. Similar to the series of word embeddings used when employing transformers to text in NLP, the ViT model represents an input image as a set of image patches and predicts class labels for the image. The patches or visual tokens are incorporated into a series of fixed-dimension encoded vectors. The encoded vector and the position of a patch in the image are both embedded and sent into the transformer encoder network for classification.

ViT’s self-attention layer makes it feasible to globally embed information throughout the entire image. To recreate the structure of the image, the model also learns from training data to encode the relative locations of the image patches. The transformer encoder includes:

- Concatenating all attention outputs linearly to the appropriate dimensions is the Multi-Head Self Attention Layer (MSA). Several ‘attention heads’ aid in the training of an image’s local and global dependencies.
- A two-layer Multi-Layer Perceptrons (MLP) Layer with a Gaussian Error Linear Unit (GELU).
- Layer Norm (LN), which does not introduce any additional dependencies between the training images, is applied before each block. As a result, training time and performance as a whole are improved.

Additionally, after every block, residual connections are added since they enable direct component flow across the network without passing through non-linear activations. The MLP layer implements the classification head in the context of image classification. It does this by using one hidden layer during the pre-training phase and one linear layer during the fine-tuning phase.

### 3.2 Data-Efficient Image Transformer (DEIT)

While ViT, needs to be pre-trained with hundreds of millions of images using external data, it does not generalize well when trained on insufficient amounts of data. DEIT, was proposed to address this issue with mostly the same architecture as ViT, with an introduction of a teacher-student strategy with distillation token.

DEIT adds a distillation token that works similarly to the class token. Ahead of the first transformer block, this token is added to the initial embeddings, patches, and class tokens. It is produced by the network and engages in self-attention interaction with the other embeddings. To allow the model to learn from the teacher’s output while staying complimentary to the class token that is tasked with reproducing the real label  $y$ , the distillation token’s goal is to replicate the hard label  $y^t$  provided by the teacher network. The distillation embedding is also learned by the transformer network by back-propagation, just like the class embedding.

### 3.3 Swin Transformer

Microsoft introduced the Swin Transformer, which incorporated hierarchical feature maps and shifted window attention, to remedy the shortcomings of the original ViT. The intermediate tensors created from each succeeding layer, combined from layer to layer, are what make up hierarchical feature maps. This essentially reduces the spatial dimension of feature maps from one layer to the next. The Swin Transformer can be used in cases where fine-grained prediction is necessary, like semantic segmentation, thanks to these hierarchical feature maps.

The Window MSA (W-MSA) and Shifted Window MSA (SW-MSA) modules used in Swin Transformer replace the function of the regular multi-head self-attention (MSA) module used in ViT. There are two components in the Swin Transformer block, with a normalisation layer, an attention module, followed by another normalisation layer, and an MLP layer making up each sub-unit. While the second sub-unit makes use of a Shifted Window MSA (SW-MSA) module, the first sub-unit employs a Window MSA (W-MSA) module.

### 3.4 Network Visualization

We employed a variety of CAM generating techniques, such as GradCAM, GradCAM++, ScoreCAM, etc., as modified for transformer networks by Gildenblat and contributors [2021], to visualise the attention map from the trained transformer networks. Comparisons were made between the CAMs produced using various configurations from the three models, including intermediate layers, all

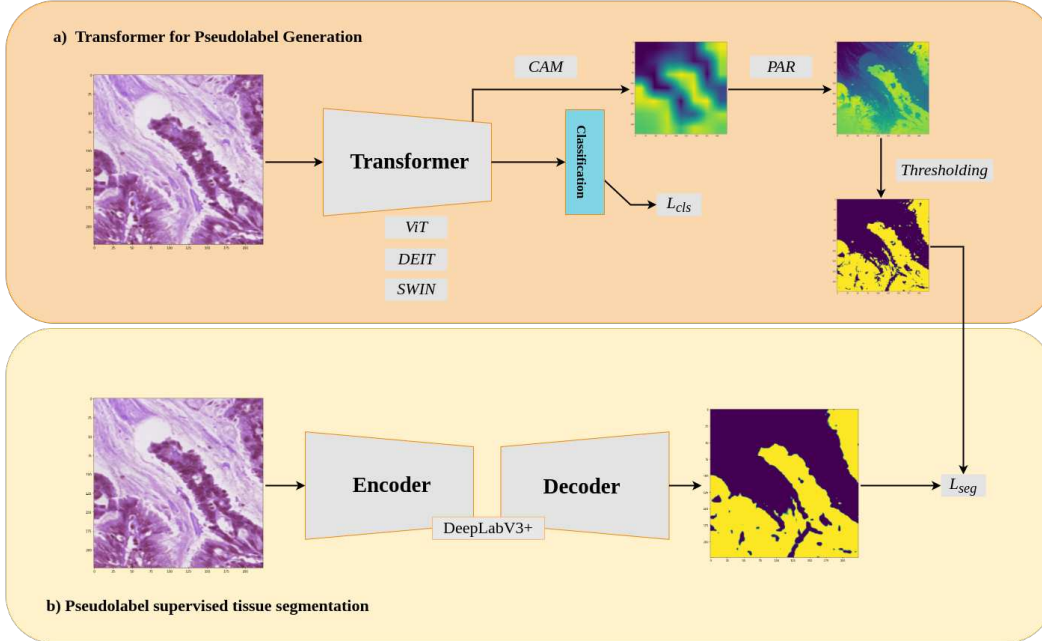


Figure 1: Proposed WSSS LUAD tissue segmentation. a) Transformer network for pseudo-label generation. b) DeepLabV3+ for semantic segmentation

layers, and final layers. For a better depiction of transformer attention, we also investigated the use of attention rollout, gradient attention rollout Gildenblat and contributors [2020], and Transformer Interpretability Beyond Attention Visualization Chefer et al. [2021]. To improve the display of the transformer attention map, we also investigated the viability of applying the Gradient-Weighted Element wise Transformer Attention Map (GETAM) Sun et al. [2021b]. Additionally, GETAM was used to create an end-to-end WSSS technique.

To determine the best performing methodology, the CAMs produced using the various approaches were compared to the images in the validation set. The best CAM generating methods were then optimised further utilising Pixel Adaptive Refinement (PAR) and thresholding for a pseudo-label generation.

### 3.5 Pixel-Adaptive Refinement

As seen in figure 2, the initial pseudo labels are generally imprecise and locally inconsistent, meaning that adjacent pixels that have a similar low-level representation to the image might not have the same interpretation. Dense CRF Krähenbühl and Koltun [2011] is used to further the initial pseudo labels in Zhang et al. [2020] Zhang et al. [2021b] in order to maintain the local consistency. To accommodate for CRF’s high execution times, which make it unsuitable for end-to-end training, Ru et al. [2022] include RGB and spatial information to define the low-level pairwise affinity and build the suggested Pixel-Adaptive Refinement module (PAR). Given the input image  $I \in \mathbb{R}^{h \times w \times 3}$ , for the pixel at position (i, j) and (k, l), the RGB and spatial pairwise components in PAR are constructed, inspired by Araslanov and Roth [2020], which uses the pixel-adaptive convolution Su et al. [2019] to extract local RGB information for refining and are defined as:

$$k_{rgb}^{ij,kl} = - \left( \frac{|I_{jk} - I_{kl}|}{w_1 \sigma_{rgb}^{ij}} \right)^2 \quad k_{pos}^{ij,kl} = - \left( \frac{|P_{jk} - P_{kl}|}{w_2 \sigma_{pos}^{ij}} \right)^2 \quad (1)$$

Whereas  $I_{ij}$  and  $P_{ij}$  represent the RGB data and the spatial location of pixel (i, j), respectively, in practice the XY coordinates are employed. The terms  $\sigma_{rgb}$  and  $\sigma_{pos}$  in Eq. 1 stand for, respectively,

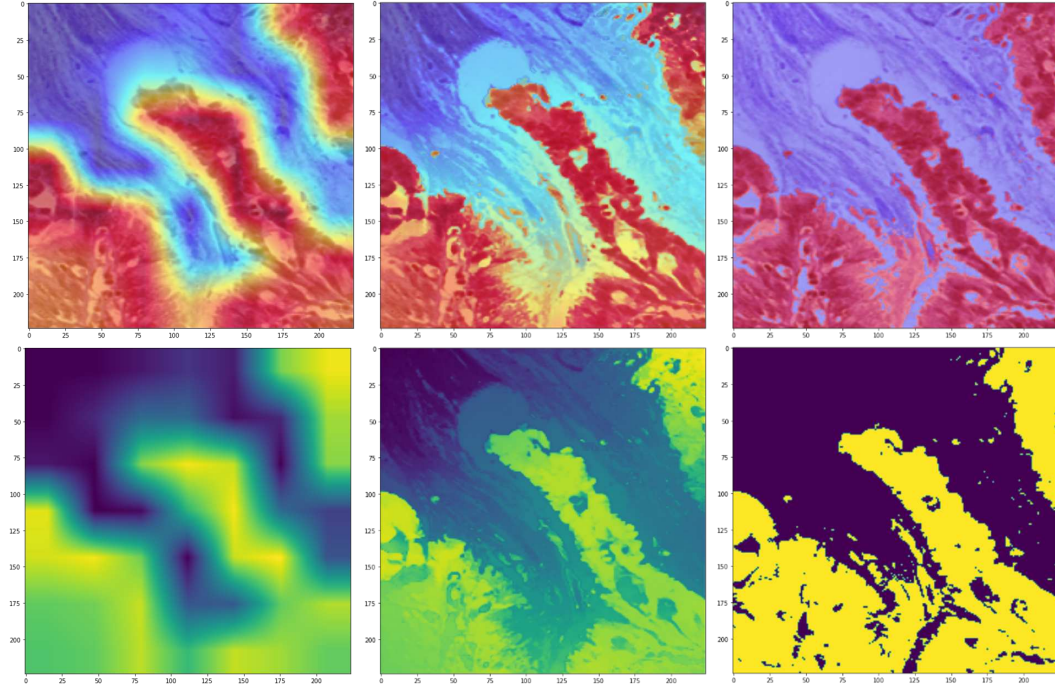


Figure 2: Refining CAMs with PAR and thresholding for pseudo-label generation. The bottom row shows the CAMs and the top row shows the CAM heat-map on the original image. The CAMs generated from the model (left) are refined using PAR (middle) and pseudo-labels are generated by thresholding the PAR refined CAM



the standard deviation of RGB and position difference. The smoothness of  $k_{rgb}$  and  $k_{pos}$  is controlled by  $w_1$  and  $w_2$ , respectively. Then, by adding  $rgb$  and  $pos$  after normalising them with softmax, the affinity kernel for PAR is created, i.e.,

$$k^{ij,kl} = \frac{\exp(k_{rgb}^{ij,kl})}{\sum_{x,y} \exp(k_{rgb}^{x,y})} + w_3 \frac{\exp(k_{pos}^{ij,kl})}{\sum_{x,y} \exp(k_{pos}^{x,y})} \quad (2)$$

Where  $N(i, j)$  is the neighbour set from which  $(x, y)$  is sampled, and  $w_3$  modifies the position term's importance. Multiple iterations of the refinement are carried out and in each  $t$  iteration, for  $CAM$   $M \in \mathbb{R}^{h \times w \times c}$ , we have,

$$M_t^{i,j,c} = \sum_{(k,l) \in N(i,j)} k^{ij,kl} M_{t-1}^{k,l,c} \quad (3)$$

The 8-way neighbours with multiple dilation rates are defined as the neighbour pixel sets  $N(\cdot)$  by Ru et al. [2022] following Araslanov and Roth [2020]. A design like this ensures training effectiveness because  $3 \times 3$  dilated convolutions make it simple to retrieve a pixel's dilated neighbours.

### 3.6 Semantic Segmentation

For semantic segmentation, we chose to use DeepLabV3+Chen et al. [2018] with ResNet50 and ResNet101 backbones, as these have shown remarkable performance for segmentation of histopathological images Han et al. [2022b] Han et al. [2022a].

## 4 Dataset

Our research focused on weakly supervised semantic segmentation of lung adenocarcinoma histopathological image, the WSSS4LUAD dataset Han et al. [2022b] (also referred to as LUAD dataset), figure 3, made available as part of the WSSS4LUAD challenge on the Grand Challenge website.

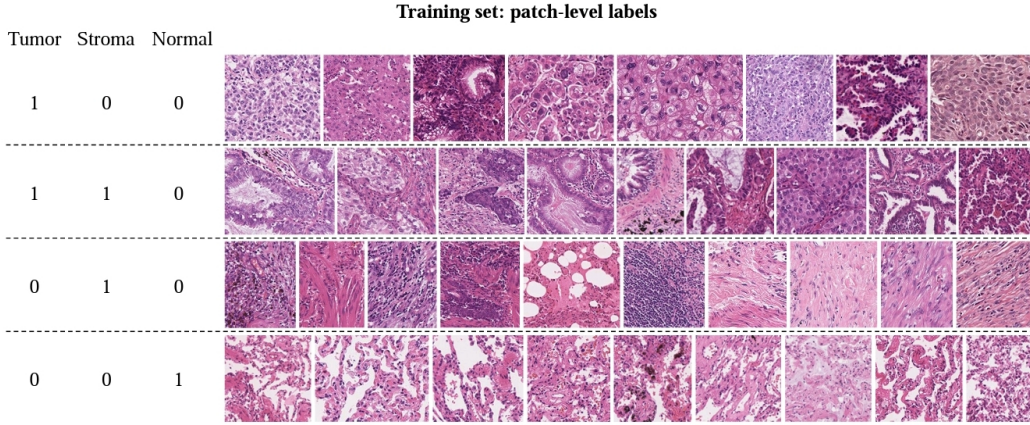


Figure 3: Images from the WSSS4LUAD Grand challenge Dataset

The training set comprises 14 WSIs from The Cancer Genome Atlas Program and 49 WSIs drawn at random from the Guangdong Provincial People's Hospital (GDPH) (TCGA). 10,091 patched images with arbitrary resolutions (between 150 - 300\*150 - 300) make up the dataset. The dataset statistics, figure 5, shows that tumor and stroma without normal ([1,1,0]) comprises most of the dataset. With just one tissue category, the labels [1,0,0], [0,0,1], and [0,1,0] have the second-highest number of combinations, however some label combinations are extremely rare or even nonexistent.

A total of 40 patches make up the validation set, with 9 large patches (about 1500-5000\*1500-5000) and 31 small patches (around 200-500\*200-500). The test set consists of 66 small patches (about 200-500\*200-500) and 14 large patches (approximately 1500-5000\*1500-5000) for a total of 80 patches.

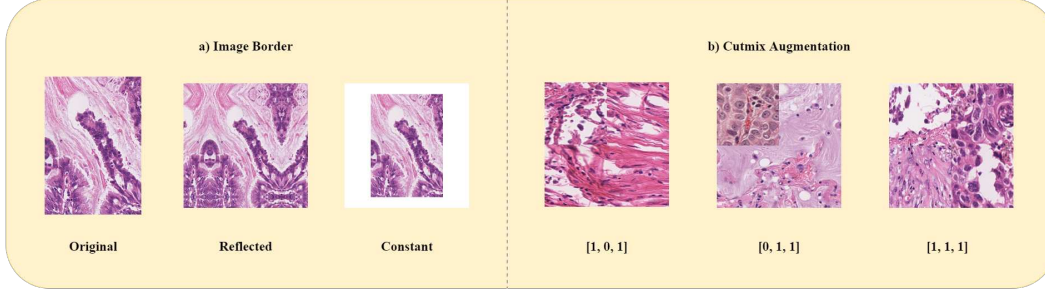


Figure 4: Data Augmentations. a) Reflected and Constant border, b) Cutmix Augmentation

#### 4.1 Data Augmentation

We used data augmentation to compensate for the training images' arbitrary resolutions and the dataset's unbalanced data distributions. We created two sets of images—one with a constant white border and the other with reflected pixels—to account for the adjustments that the picture scaling would entail. The borders were created to convert the image to 300\*300 resolution. The augmented images were then resized to a resolution of 224\*224 pixels and cut-mix augmentation was also used to address the severely constrained data labels from nonexistent to rare combination of image labels as described in figure 5. Figure 4 shows the resultant images from the augmentation. On top of the aforementioned augmentations, on run-time, image rotations and horizontal/vertical flips were also employed.

### 5 Experiments

In this section, we provide the numerous experiments carried out for a thorough assessment of our strategy for using vision transformer networks for the generation of pseudo-labels and how effective it is for tissue semantic segmentation.

#### 5.1 Networks and CAM

We started working on the ViT network with two configurations: one with 16\*16 patches and the other with 32\*32 patches, and an input image size of 224\*224. Both networks were trained using only the original dataset and the supplemented dataset for "multi-label" and "multi-class" classification. We started working on DEiT and Swin transformer networks after determining the best-performing network configuration from ViT. We trained both the networks for "multi-class" and "multi-label" classification. After the models had been trained for classification, we looked at the CAMs produced by cutting-edge methods like GradCAM, GradCAM++, ScoreCAM, etc. as they had been modified for neural networks in Gildenblat and contributors [2021].

As is evident from figure 6, the multi-class classification model outperformed the multi-label classification models in terms of CAM production, despite the multi-label task's higher classification accuracy. Following that, we also see that utilising cutmix augmentation to forcibly expand the dataset to include additional image classes does not perform as well as it did for the WSSS4LUAD grand competition competitors using the CNN approach. To cover more of the class-specific sections precisely, we employed the gradient attention rollout Gildenblat and contributors [2020] Chefer et al. [2021], but unlike natural scene images, this method did not increase the precision of the CAMs created, instead highlighting just the most discriminative parts, if any at all. As they had shown cutting-edge results on the PASCAL VOC and COCO datasets, we went on to use the framework from Sun et al. [2021b] to generate the cam. However, again in contrast to what is seen in the natural scene images, Sun et al. [2021b] did not perform any better, and the best CAMs were produced using GradCAM and ScoreCAM.

We chose the GradCAM as our default approach for generating the CAMs from the networks to avoid the lengthy processing time for creating the ScoreCAM. Now, in contrast to natural scene photos where the off-the-shelf models can be utilised to build saliency maps to constrain and refine the CAMs, we are unable to efficiently generate saliency maps for histopathology images to refine the CAMs. As a result, to improve the CAMs produced by the transformer networks, we employ PAR, which makes use of the original image to produce high-quality CAMs from the initial GradCAM, as seen in figure 2. We threshold the original image from the corrected initial pseudo labels to remove the white background information. Following the removal of the backdrop, these CAMs are thresholded to only keep the portions of the CAM that are pertinent to each class.



The default PAR parameters were set to Ru et al. [2022] with the weight factors ( $w_1, w_2, w_3$ ) and dilation rates for nearby pixels at (0.3, 0.3, 0.01) and [1, 2, 4, 8, 12, 24] respectively.

## 5.2 Semantic Segmentation

The ResNet50 and ResNet101 backbones were used to train the pseudo-label supervised semantic segmentation model (DeepLabV3+). Both models were trained on images with a single label as well as images with multiple labels. According to our findings, the model with ResNet backbone trained on an image with only one label outperformed all other configurations.

## 5.3 Training

The classification models were trained with Adam optimizer with the learning rate set to 0.0001 for 32\*32 models and 0.00001 for 16\*16 models. For multi-class classification, the cross-entropy loss was used and for multi-label classification, the multi-label soft margin loss was used during training. The segmentation models were trained with the AdamW optimizer with a learning rate of 0.00001 and a learning rate decay of 0.1. A Sum of Two loss functions, dice loss and Jaccard loss, was used to train the segmentation models.

## 6 Environment

The project leverages Google’s Colab Pro web IDE, utilizing the PyTorch library for Python with 12GB RAM and 15GB of cloud storage. Security and confidentiality are not a concern given the public nature of the data used.

## 7 Results

| Data | mIoU | Tumor | Stroma | Normal |
|------|------|-------|--------|--------|
| Val  | 0.58 | 0.58  | 0.38   | 0.78   |
|      | 0.77 | 0.76  |        | 0.78   |
| Test | 0.63 | 0.53  | 0.42   | 0.94   |
|      | 0.90 | 0.86  |        | 0.94   |

Table 1: Pseudo-label supervised Tissue Segmentation

In this section, we report the quantitative analysis of pseudo-label supervised tissue segmentation using the Jaccard Index (IoU). Table 1 shows the segmentation performance on the validation and test datasets. The model struggles at distinguishing the tumour and tumour-related stroma regions. However, the model performs significantly better at distinguishing the normal tissue from the tumour as well as tumor related stroma for both validation and test sets. We get a IoU score of 0.78 and 0.94 for segmentation of normal tissue for validation and test sets, respectively, better than the best score of 0.89 for the test set as reported in Han et al. [2022b]. As our model struggles with separating the tumour and stroma regions, taking both these regions as one (cancer region) and then computing the segmentation performance, we get the mIoU scores of 0.77 and 0.90 for validation and test sets respectively.

## 8 Conclusion and Future work

In this project, we present a transformer-based approach for weakly supervised segmentation given image-level labels for histopathological images. This is the first application of the vision transformer-based approach, as traditionally, convolutional neural network-based approaches have been used for WSSS in histopathological images. Transformer-based approaches have just recently started gaining more *attention* for WSSS in natural scene images. However, replicating the work on natural images on the LUAD dataset confirms the findings from Chan et al. [2021] that the methods developed for natural scene images do not particularly scale well for images from different domains. We were able to achieve a mIoU score of 0.86 and 0.94 for combined tumour-stroma and normal images respectively for the test set. From the performance numbers, we can confirm that our approach is feasible for WSSS in histopathology. Although struggling to separate the tumour and stroma regions,

our approach achieves good results in separating cancer and its associated tissues from normal tissues, resulting in the top 5 best results from the best quantitative results from Han et al. [2022b].

Further work is required to address the weak segmentation performance in the tumour vs stroma regions. As demonstrated in table 1, our method struggles with the separation of tumour regions from the tumour-associated stroma region as compared to the normal region from the two. To address these issues, further work towards aggregating more balanced data from much wider histopathological image sources is desired. Another consideration is the augmentation techniques other than cutmix augmentation as we observed from our work that cutmix augmentation is not the best augmentation method to attempt to balance the dataset for the missing images for a transformer-based approach.

Additionally, as this project primarily focused on determining the feasibility of application of the transformer-based approach for WSSS on the LUAD dataset, which we have observed to be the case, further work into fine-tuning and exploring other variations of vision transformer and CAM generation and refining methods for generating better pseudo-labels should help improve the segmentation performance where our approach is struggling at the moment. Finally, further research in developing an end-to-end framework for WSSS using a transformer backbone is another aspect of this project to be explored in the future.

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## 9 Appendix

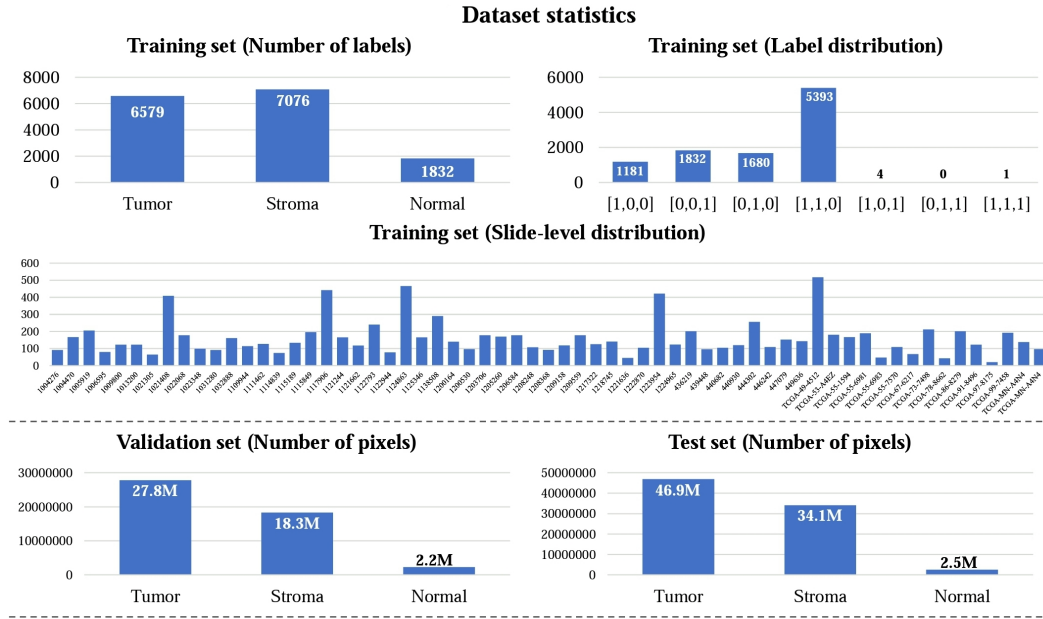


Figure 5: The number of patch-level and pixel-level labels for each tissue type in the training, validation, and test sets, respectively

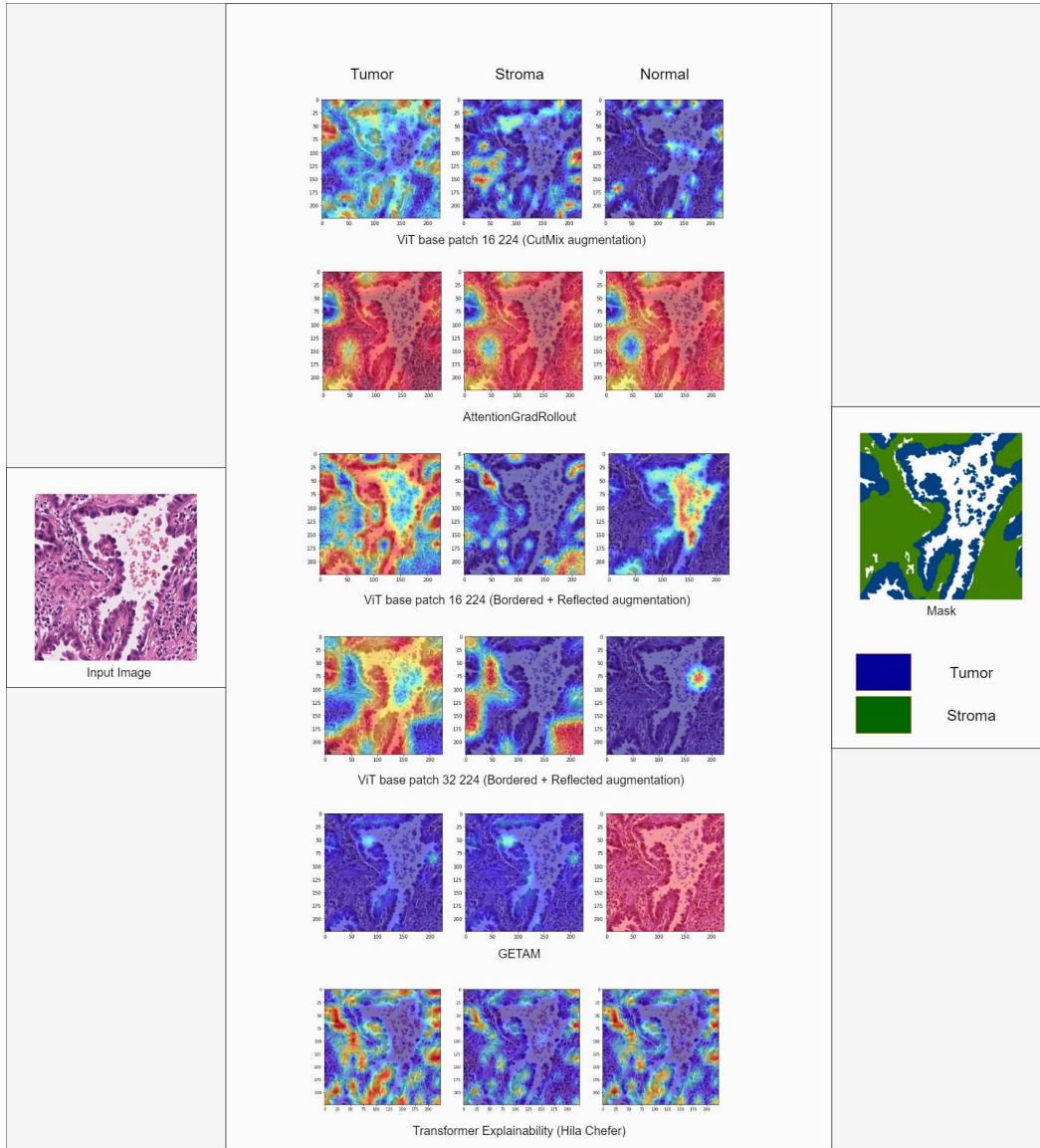


Figure 6: Comparison of all CAMs from various transformer networks and CAM generation methods